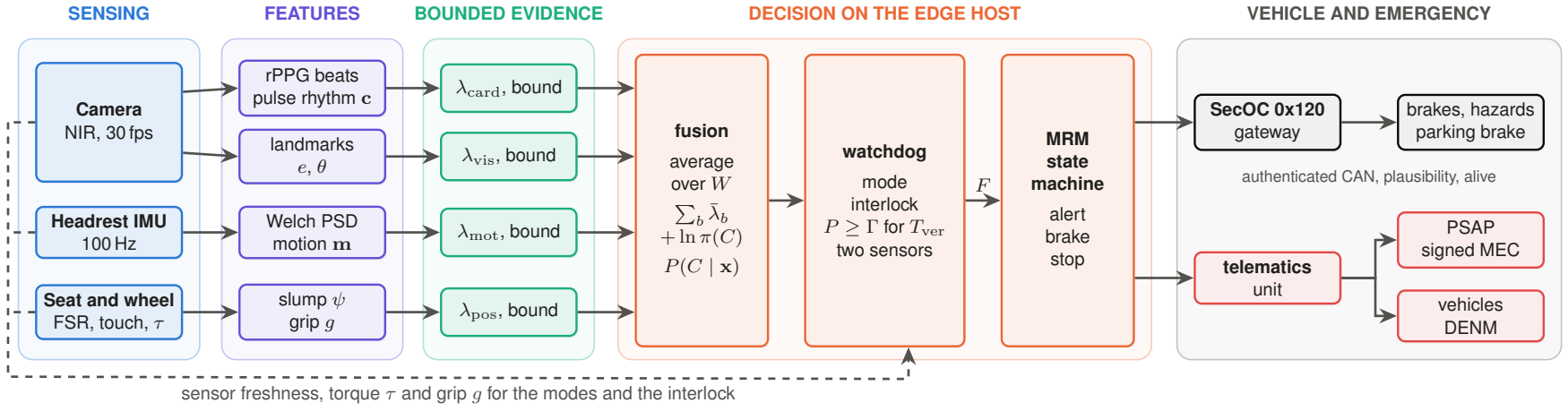

SENSING
FEATURES
BOUNDED EVIDENCE
DECISION ON THE EDGE HOST
VEHICLE AND EMERGENCY
Camera
NIR, 30 fps
Headrest IMU
100 Hz
Seat and wheel
FSR, touch, τ
rPPG beats
pulse rhythm $\mathbf{c}$
landmarks
e, θ
Welch PSD
motion $\mathbf{m}$
slump ψ
grip g
λ_{card}, bound
λ_{vis}, bound
λ_{mot}, bound
λ_{pos}, bound
fusion
average
over W
$\sum_b \bar{\lambda}_b$
$+ \ln \pi(C)$
$P(C \mid \mathbf{x})$
watchdog
mode
interlock
$P \geq \Gamma$ for T_{ver}
two sensors
F
MRM
state
machine
alert
brake
stop
SecOC 0x120
gateway
brakes, hazards
parking brake
authenticated CAN, plausibility, alive
telematics
unit
PSAP
signed MEC
vehicles
DENM
sensor freshness, torque τ and grip g for the modes and the interlock